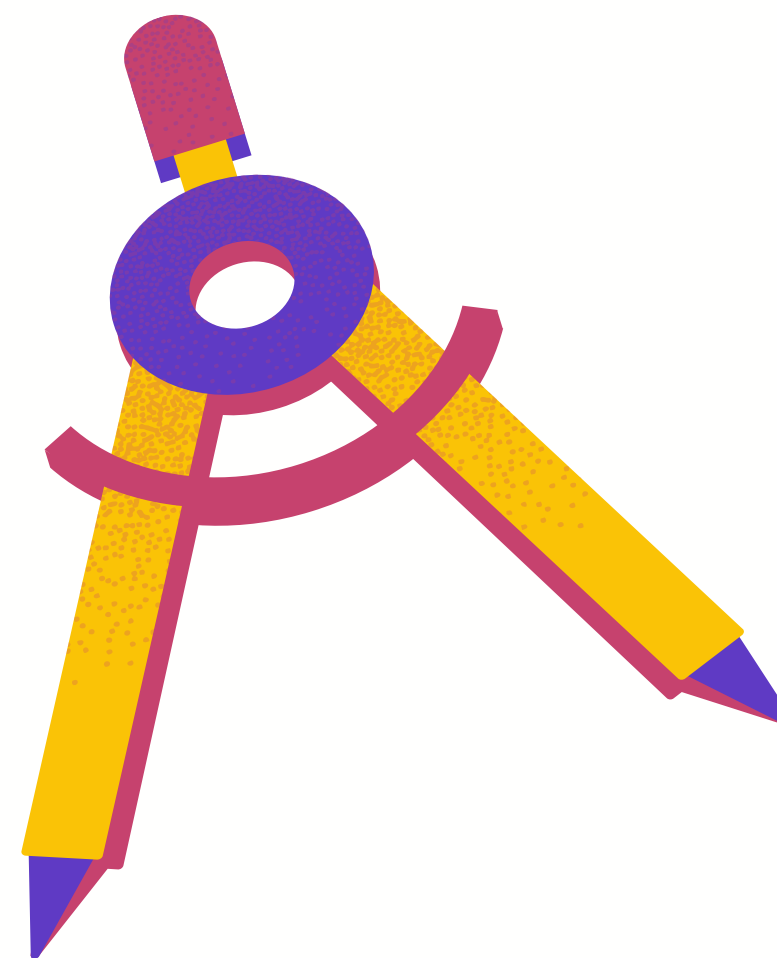


*All About*  
**ANGLES**



The image features a white background with several large, colorful geometric shapes. In the top-left corner, there is a purple triangle with a lighter purple shadow. In the top-right corner, there is a yellow triangle with a lighter yellow shadow. In the bottom-left corner, there is a red triangle with a lighter red shadow. A solid purple horizontal bar spans the entire width of the image at the bottom. The word "VOCAB" is centered in the middle of the image in a bold, red, sans-serif font with a slight shadow effect.

**VOCAB**

# Ray-



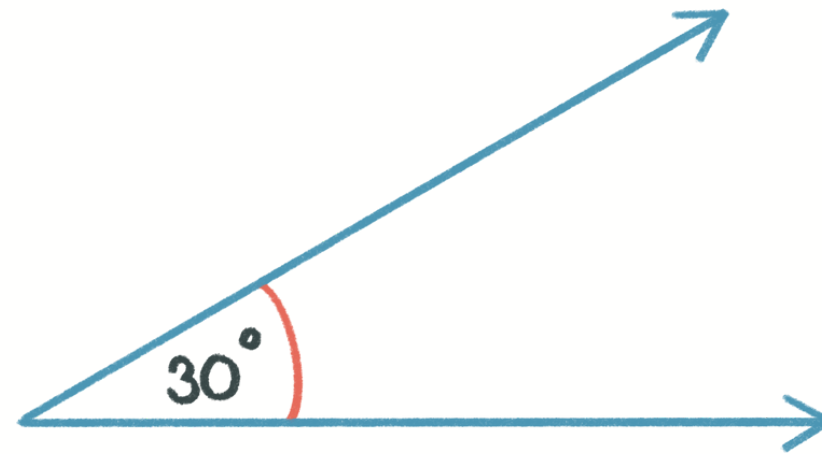
A ray is a line that moves endlessly in one direction.

# *End Point-*



An endpoint is a point that marks the end of a line segment or a ray.

# Angle -



An angle is formed when two rays share a common end point.

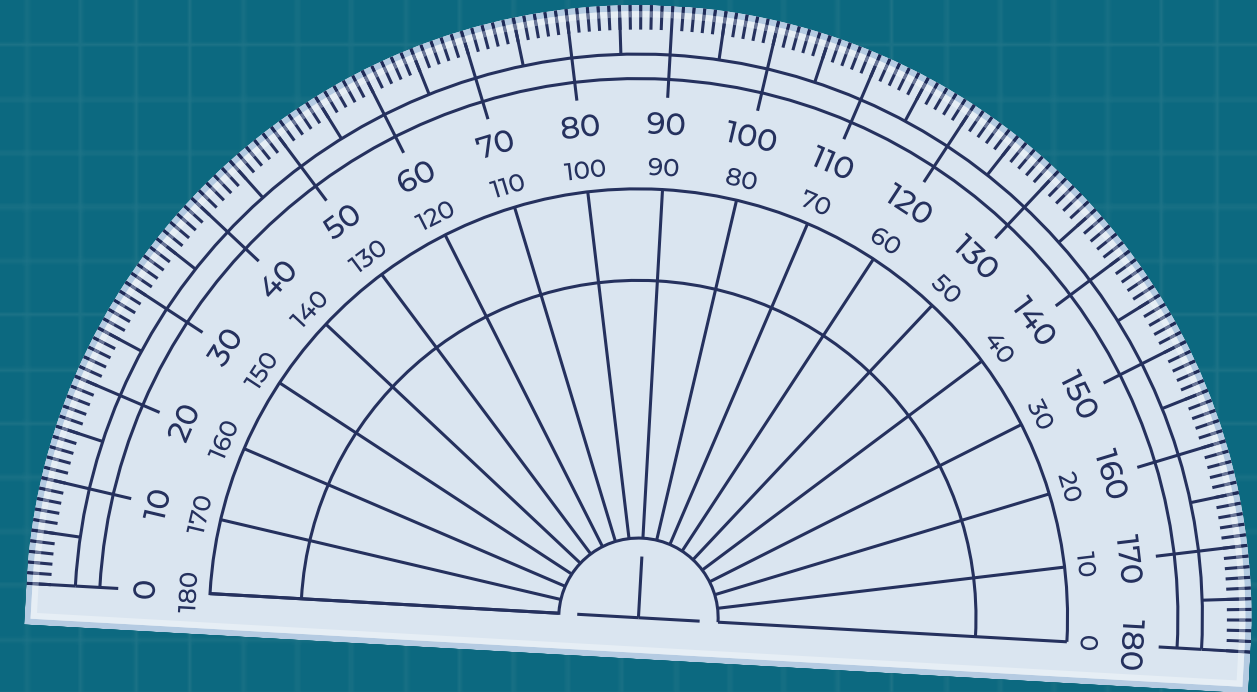


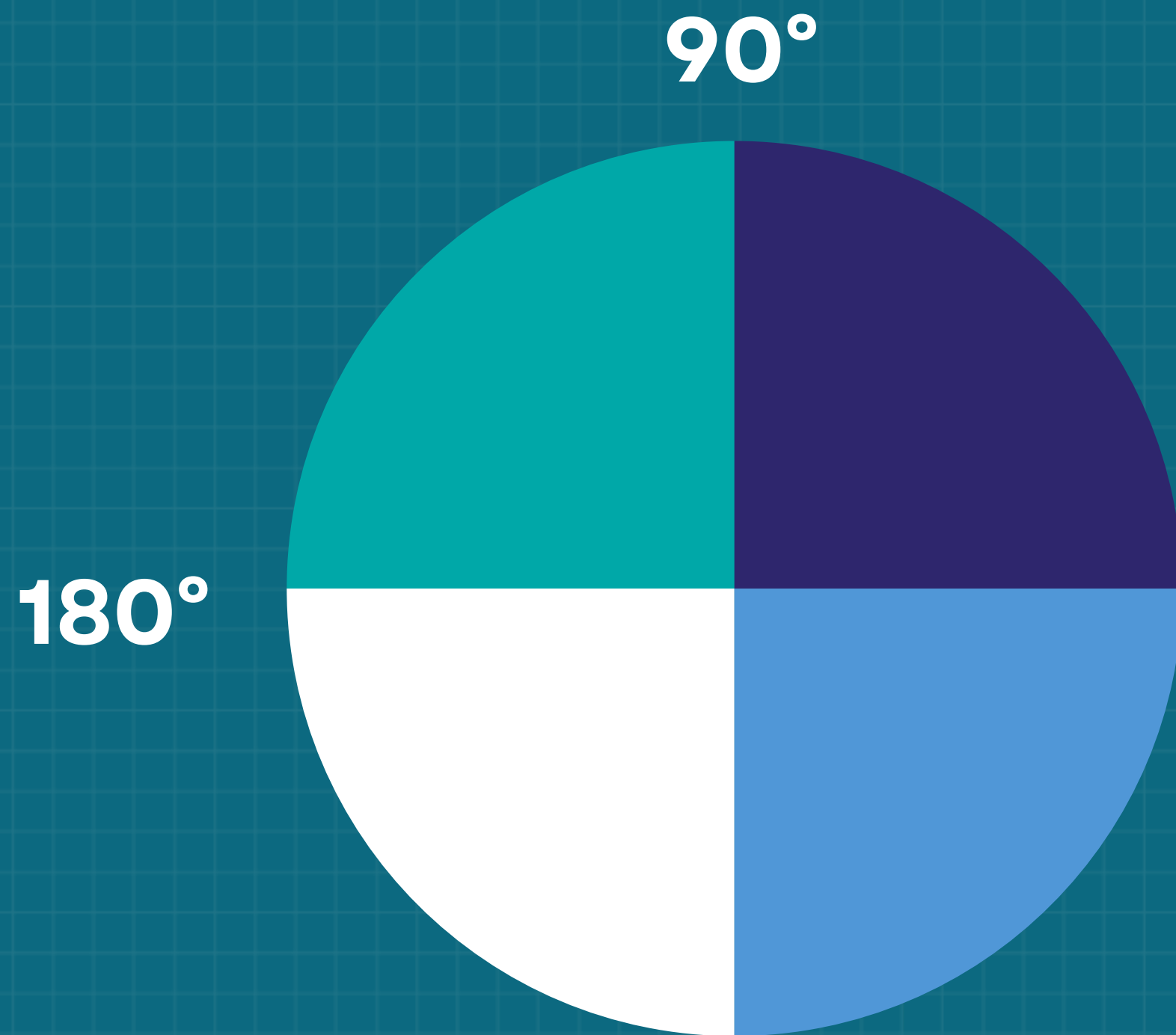
## WHAT IS AN ANGLE?

An angle is the shape formed by two rays meeting at a common end point.

# PROTRACTOR

An instrument for measuring angles. Angles are measured in degrees.

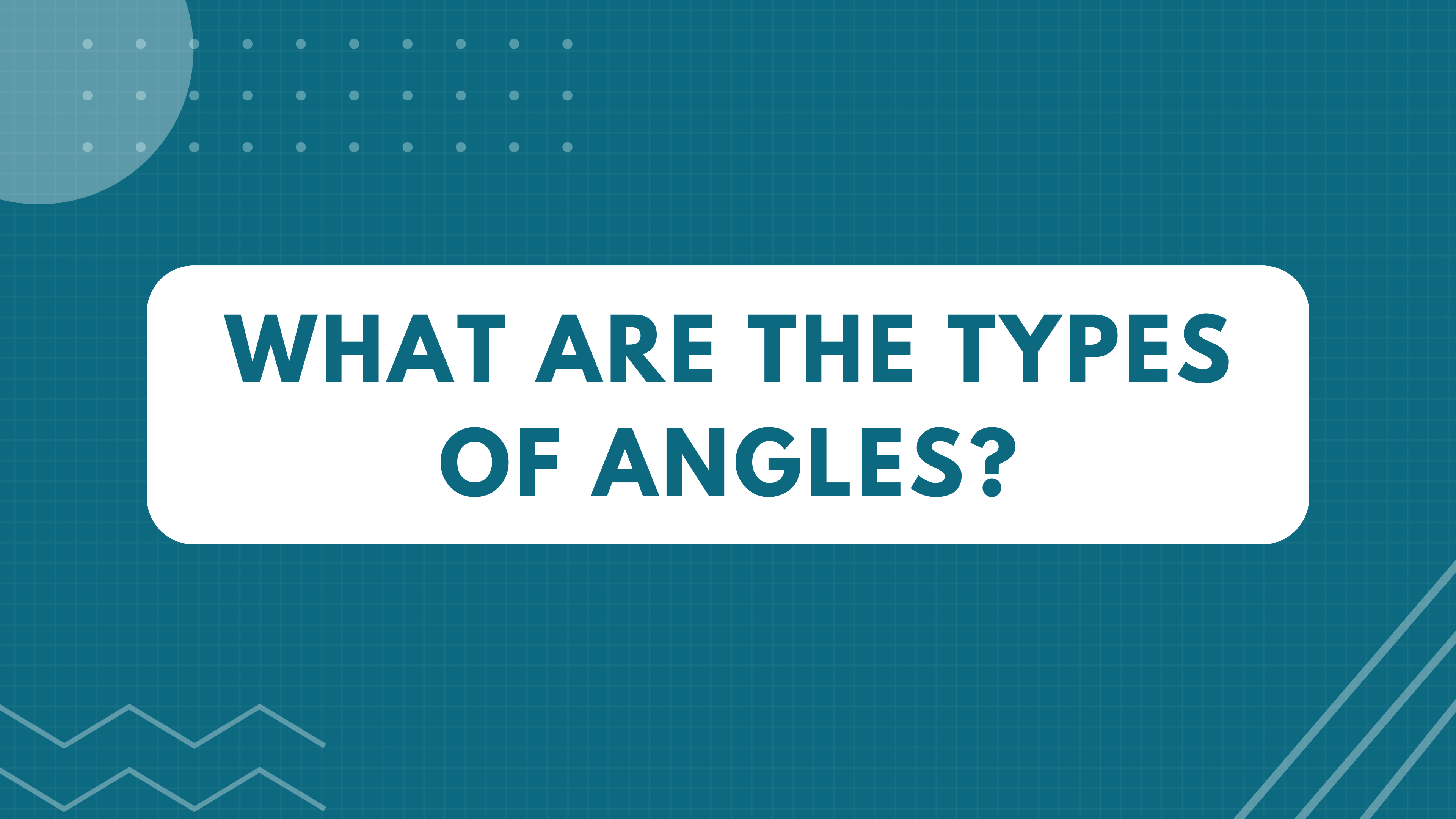




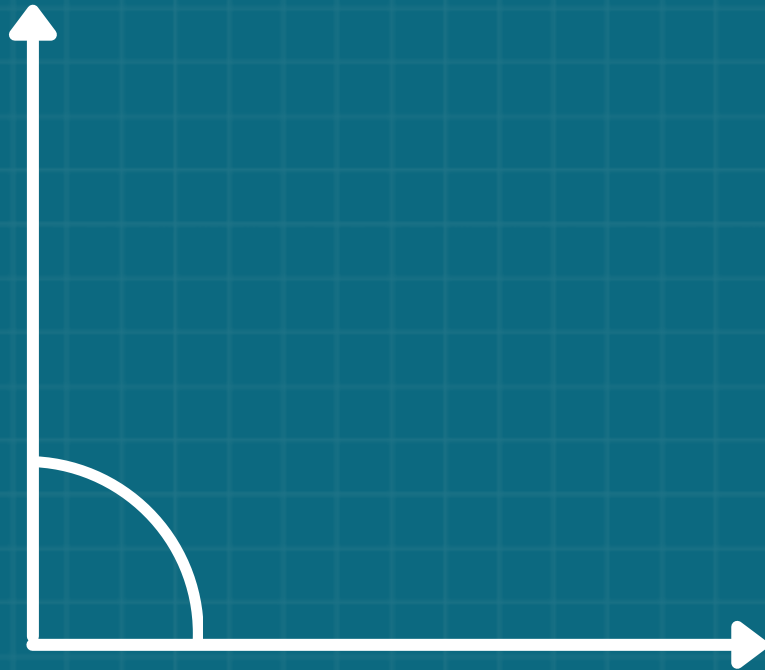
## DEGREES

Degrees are the units of measurement for angles.



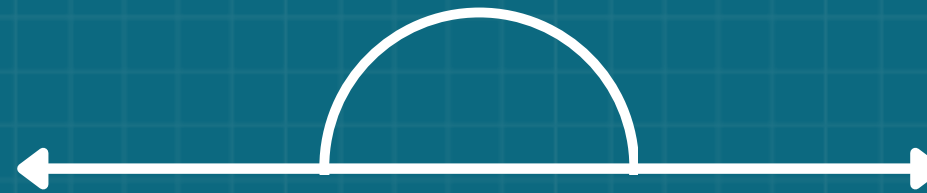


# **WHAT ARE THE TYPES OF ANGLES?**



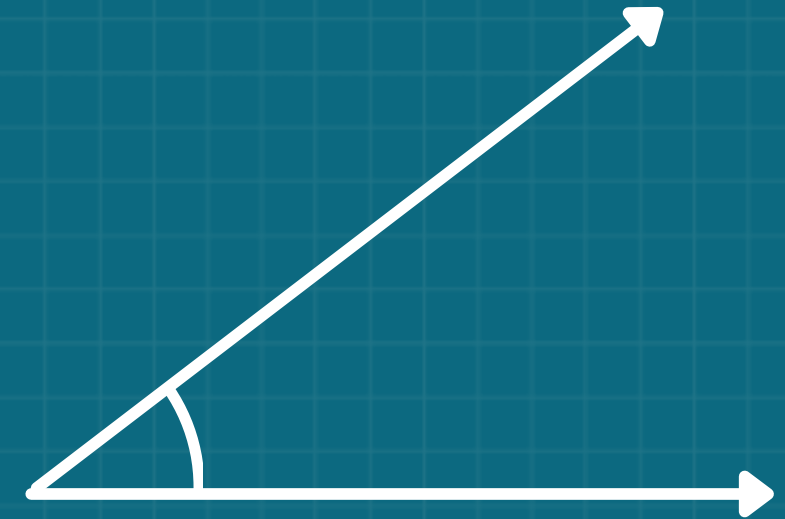
## RIGHT ANGLE

It is an angle that  
measures exactly 90  
degrees



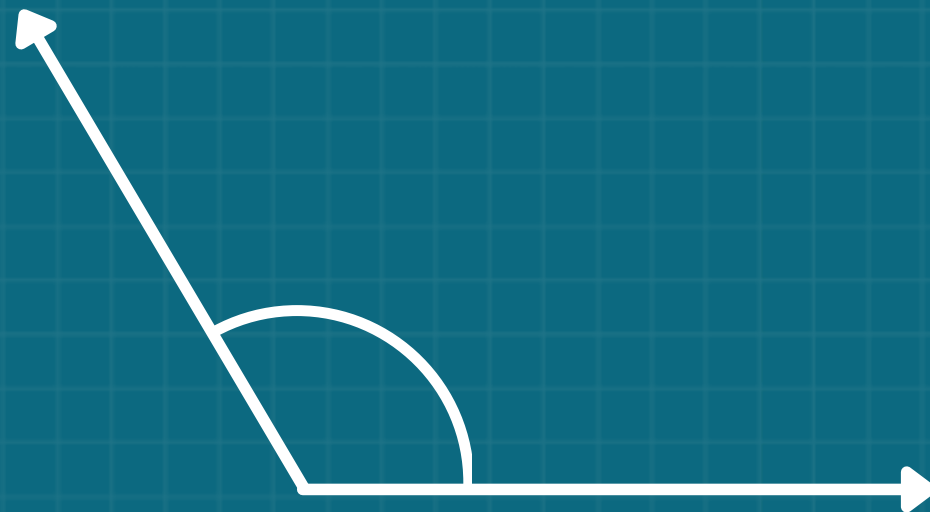
## STRAIGHT ANGLE

It is an angle that  
measures exactly 180  
degrees



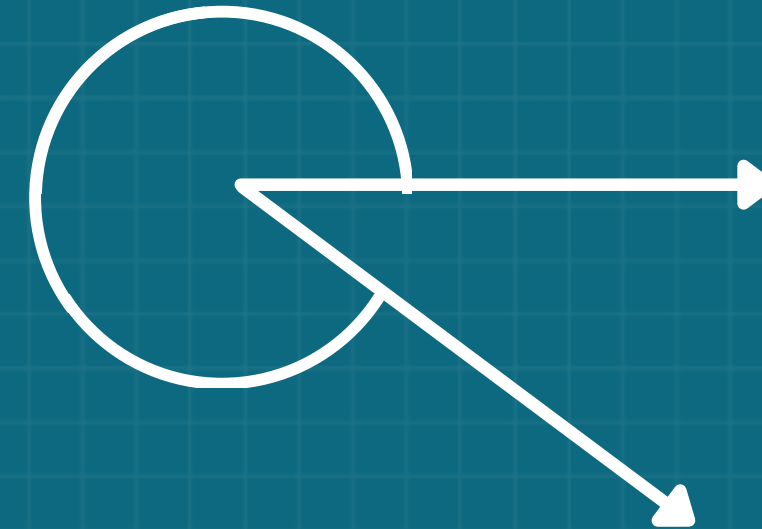
## ACUTE ANGLE

It is an angle that  
measures less than 90  
degrees



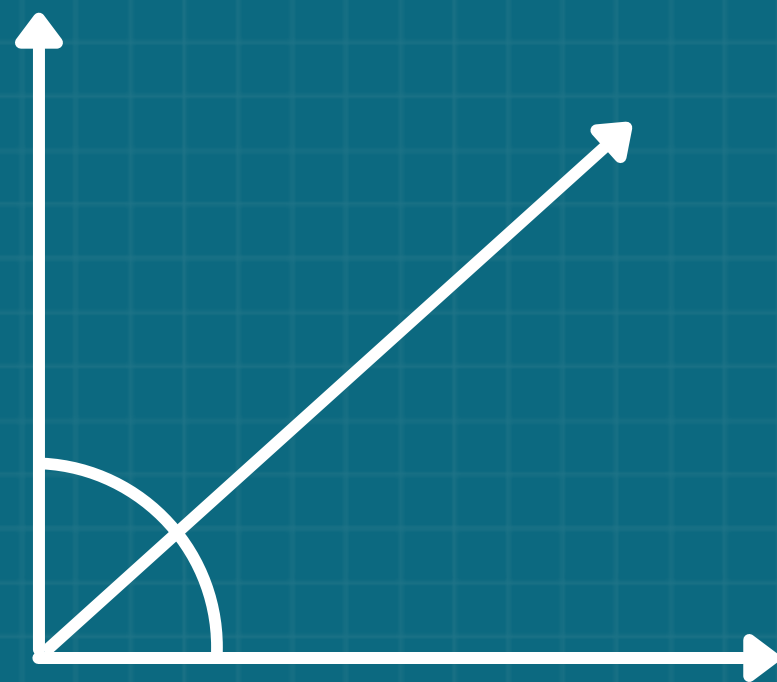
## **OBTUSE ANGLE**

It is an angle that  
measures between 90  
and 180 degrees



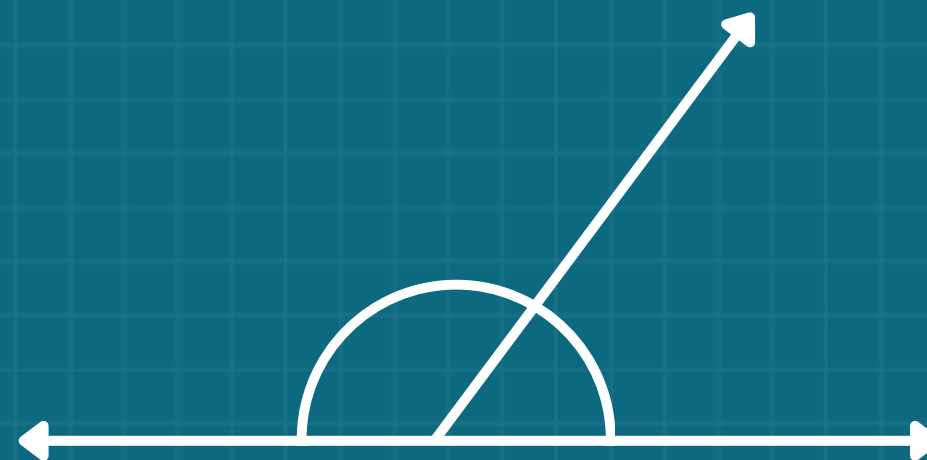
## **REFLEX ANGLE**

It is an angle that  
measures more than  
 $180^\circ$  but less than  $360^\circ$



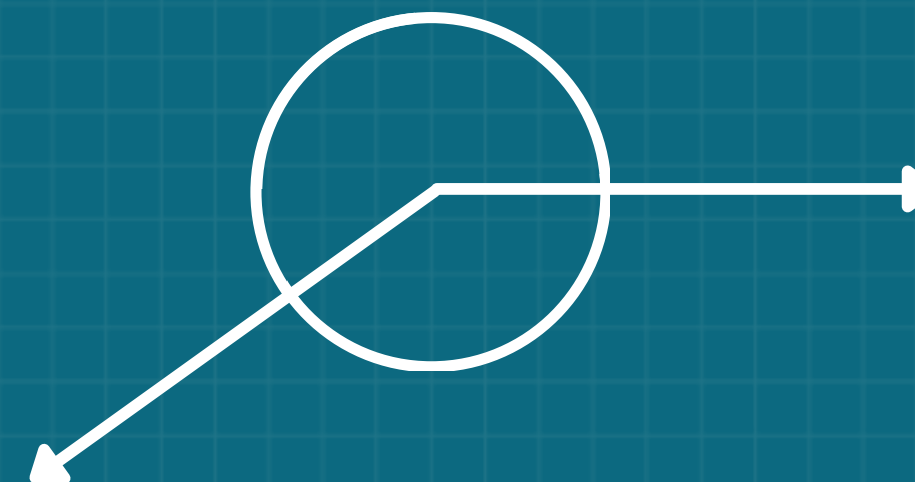
### COMPLEMENTARY ANGLE

It is a pair of angles that when added are equal to  $90^\circ$



### SUPPLEMENTARY ANGLE

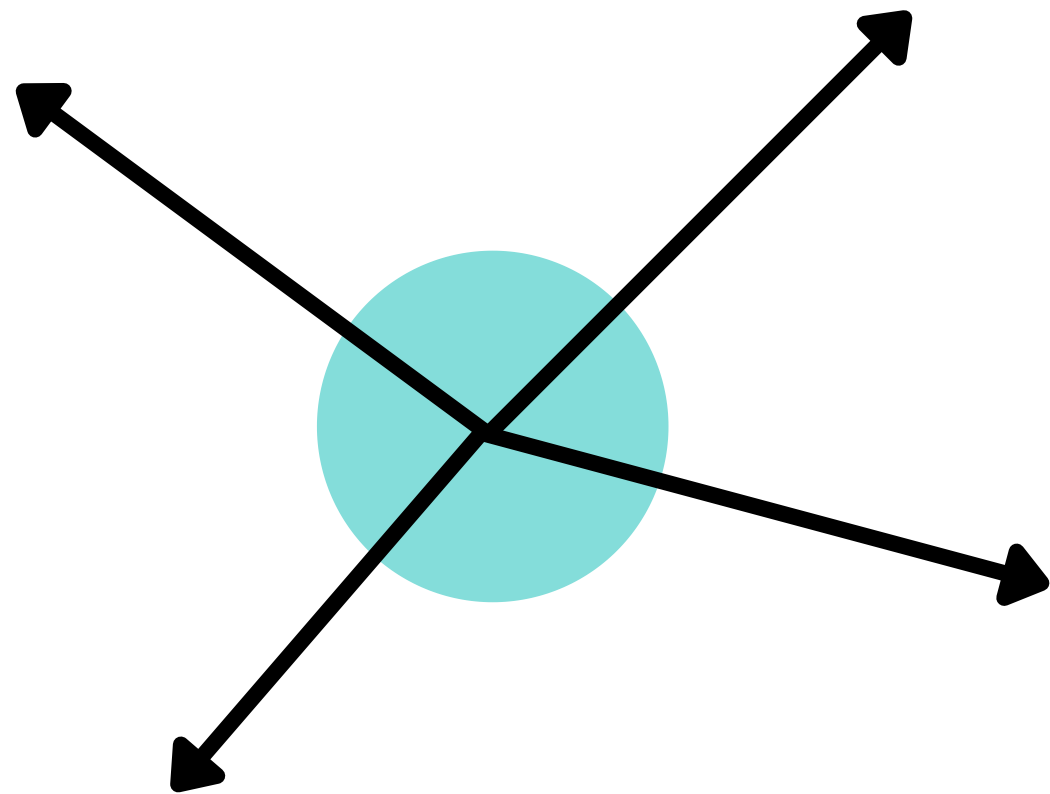
It is a pair of angles that when added are equal to  $180^\circ$



### CONJUGATE ANGLE

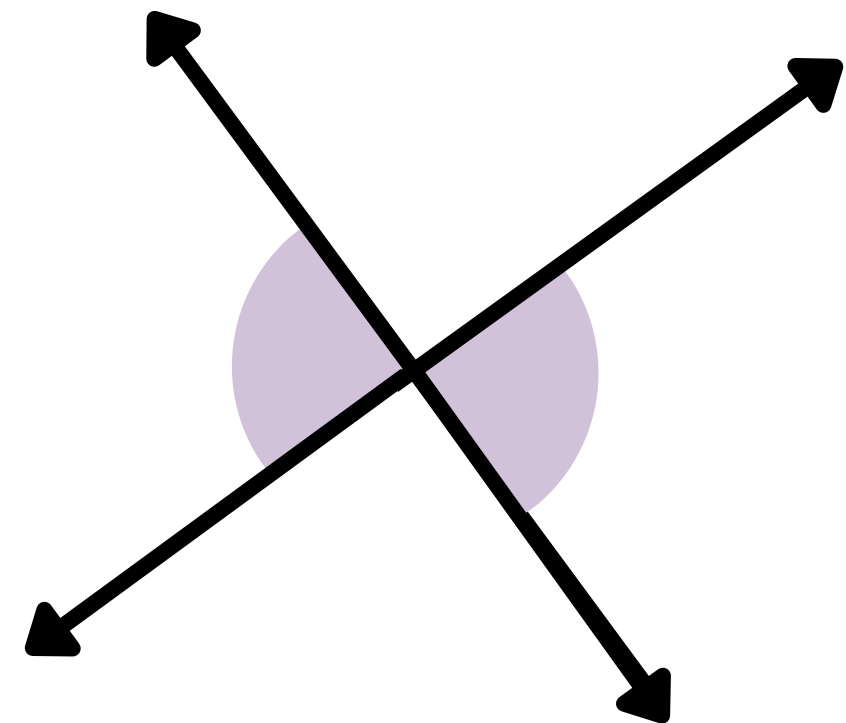
It is a pair of angles that when added are equal to  $360^\circ$

## At a Point



Angles at a point  
sum to  $360^\circ$

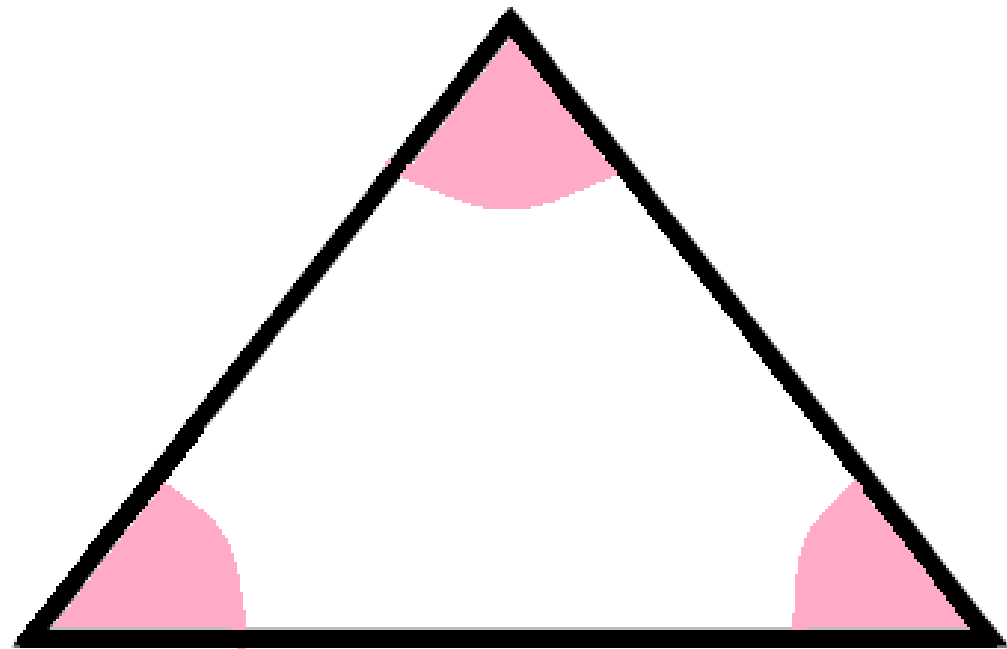
## Vertically Opposite



Vertically opposite  
angles are equal

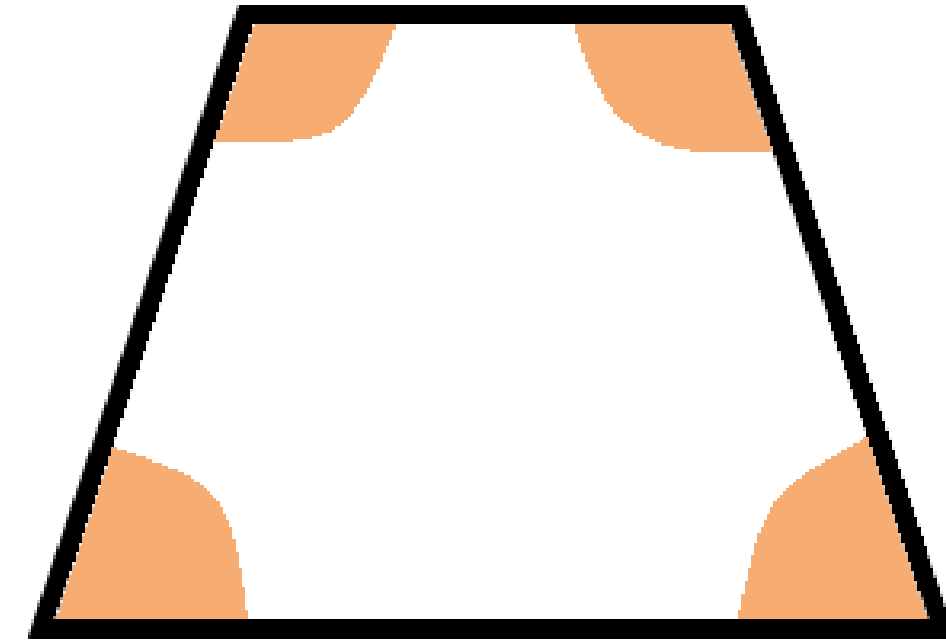


## Triangle



Angles in a triangle  
sum to  $180^\circ$

## Quadrilateral

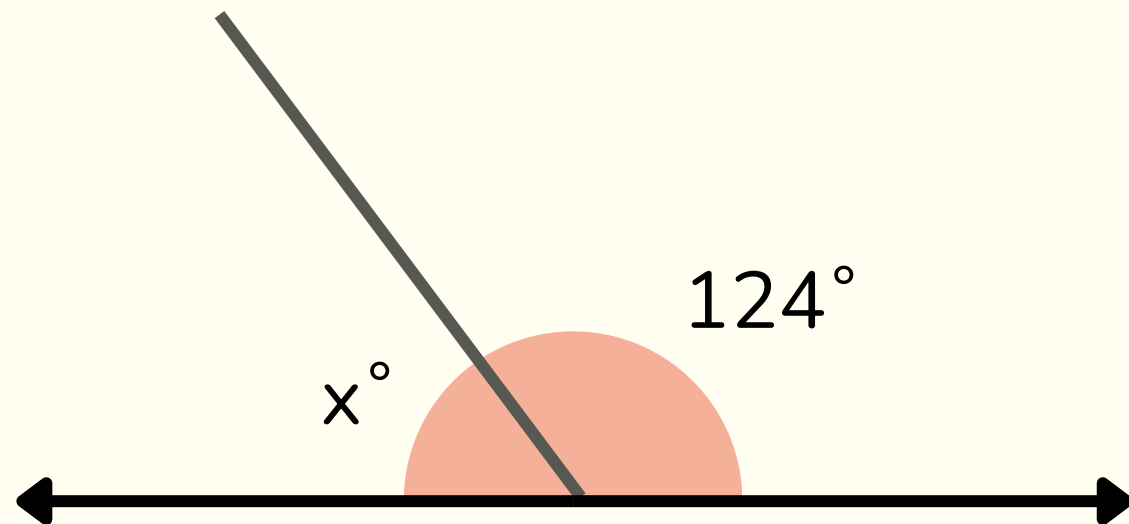


Angles in a quadrilateral  
sum to  $360^\circ$

# EXAMPLES

Find the missing angles, diagrams are not drawn accurately:

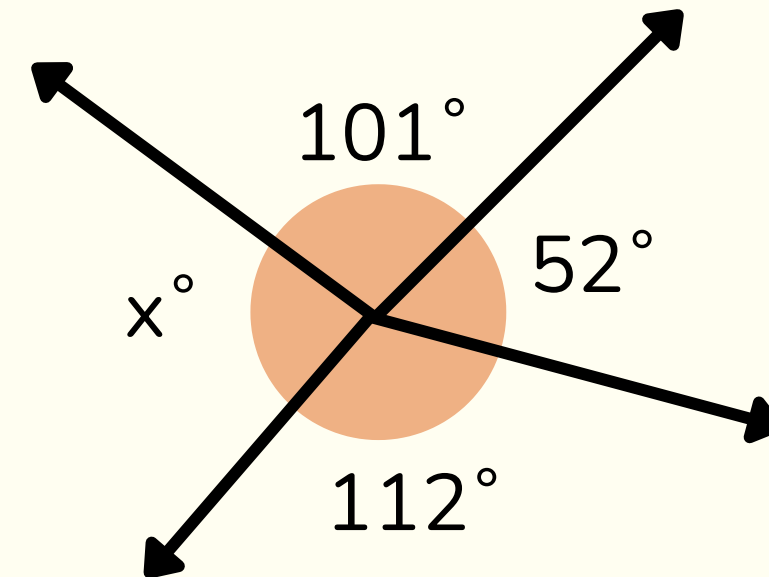
a)



$$180 - 124 = 56^\circ$$

Adjacent angles on a straight line sum to  $180^\circ$

b)



$$101 + 112 + 52 = 265^\circ$$

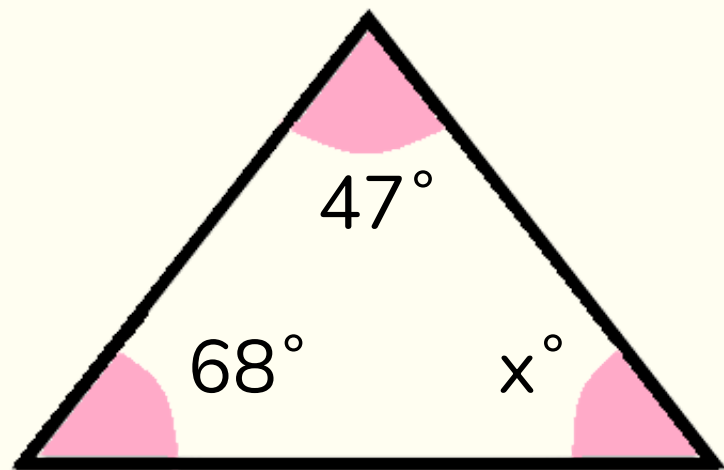
$$360 - 265 = 95^\circ$$

Angles at a point sum to  $360^\circ$

# MORE EXAMPLES

Find the missing angles, diagrams are not drawn accurately:

a)

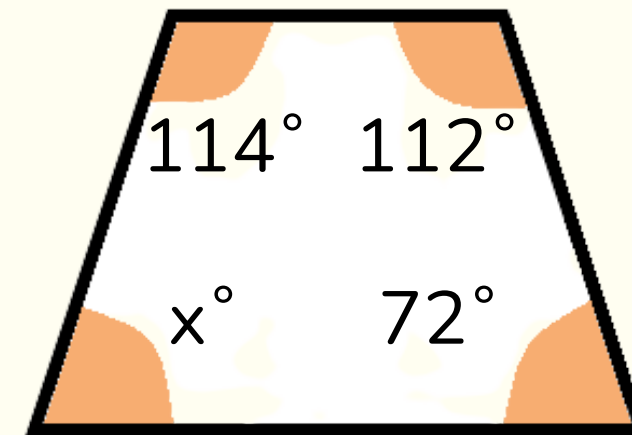


$$68 + 47 = 115^\circ$$

$$180 - 115 = 65^\circ$$

Angles in a triangle sum to  $180^\circ$

b)



$$114 + 112 + 72 = 298^\circ$$

$$360 - 298 = 62^\circ$$

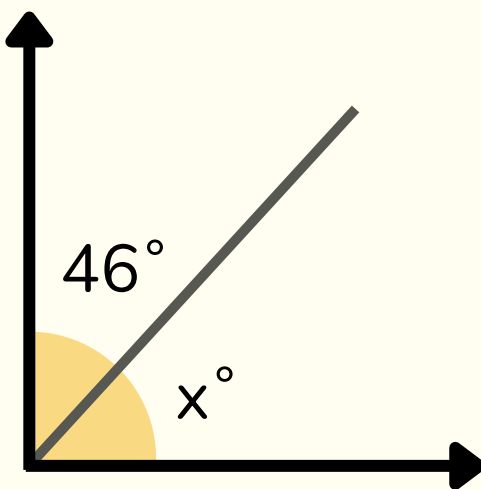
Angles in a quadrilateral sum to  $360^\circ$



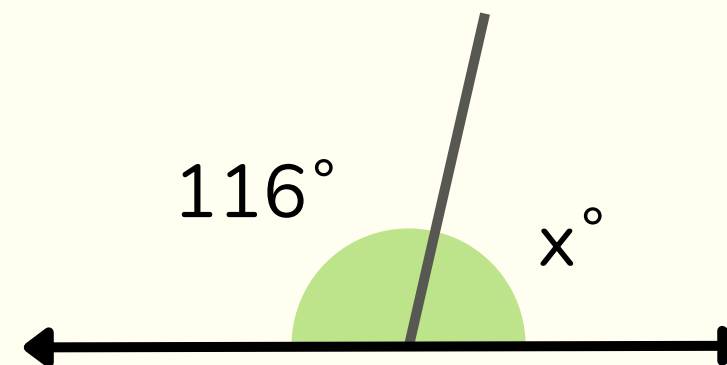
# PRACTICE QUESTIONS

Find the missing angles, diagrams are not drawn accurately:

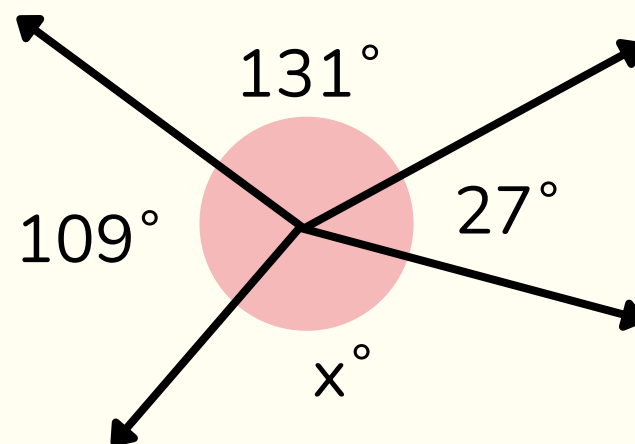
1.



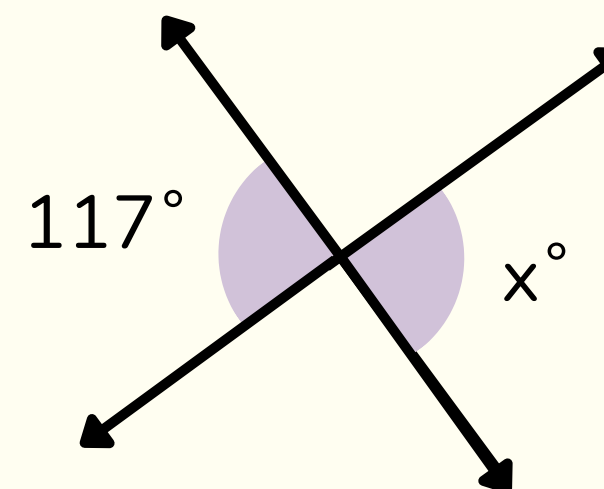
2.



3.

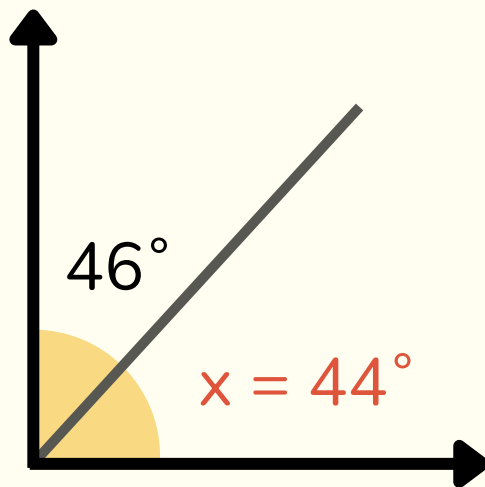


4.

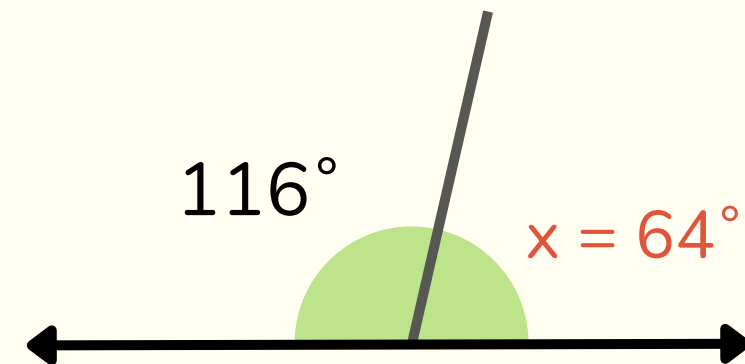


# SOLUTIONS

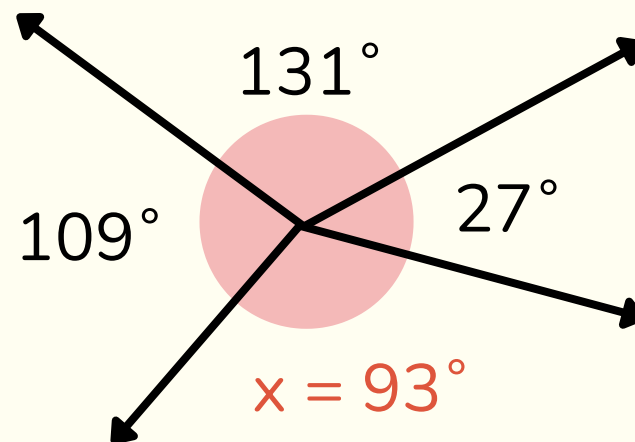
1.



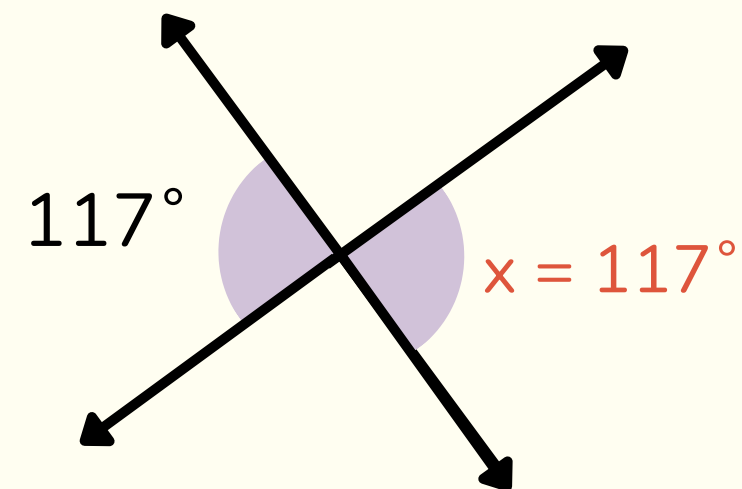
2.



3.



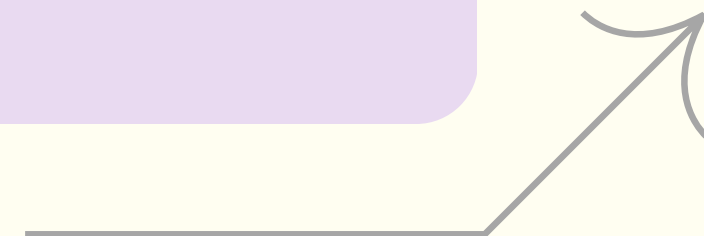
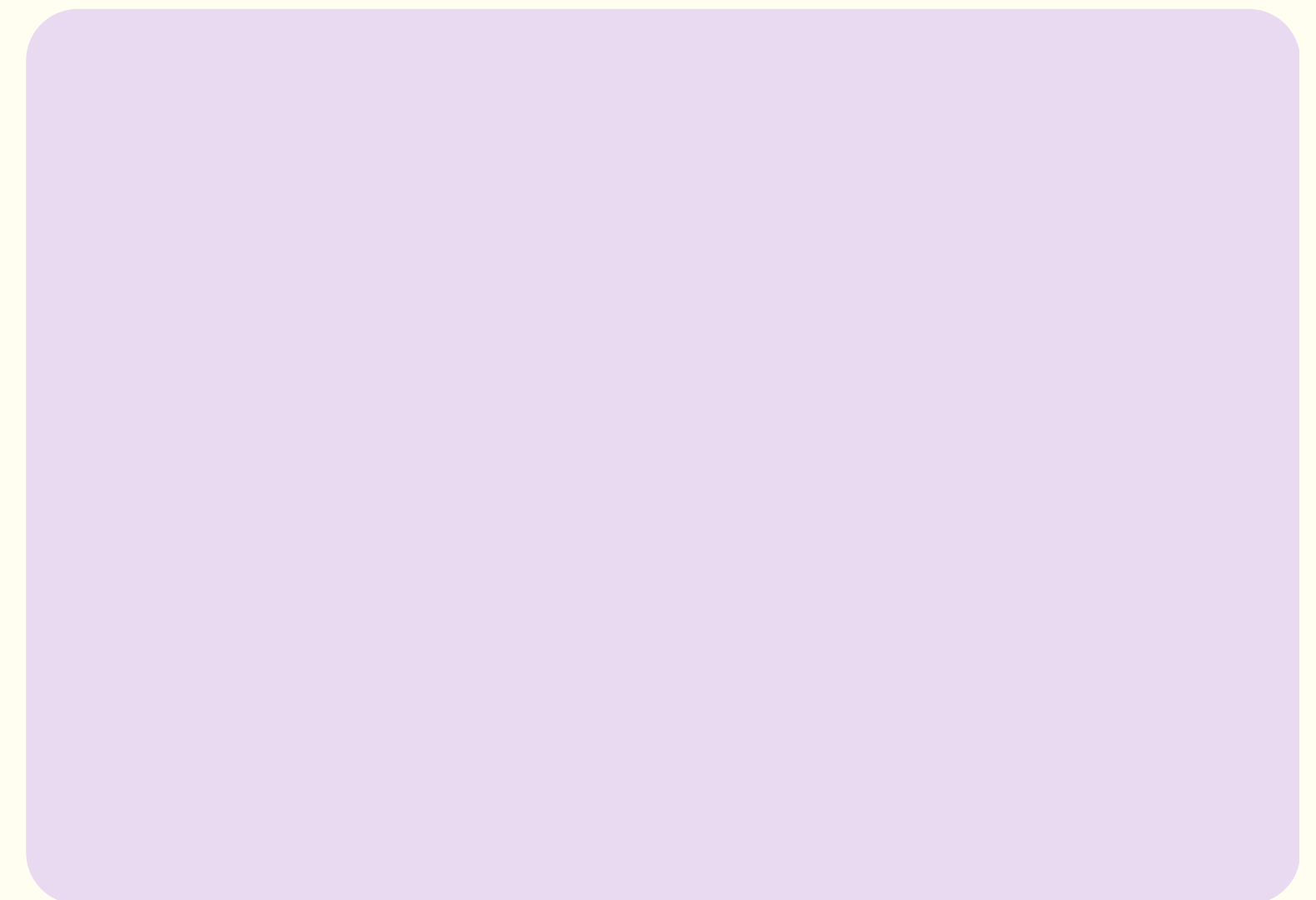
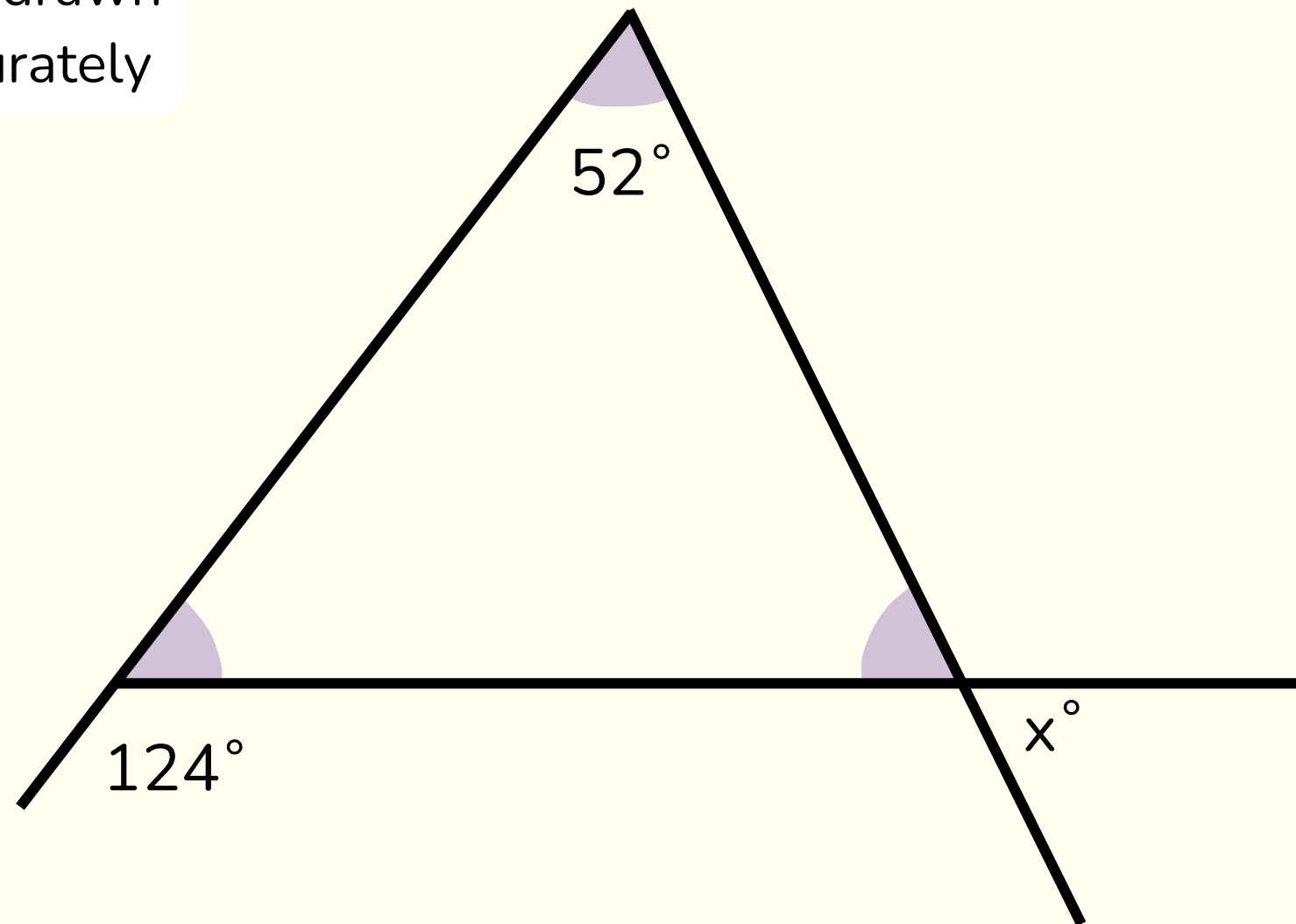
4.



# COMPLEX ANGLE PROBLEMS

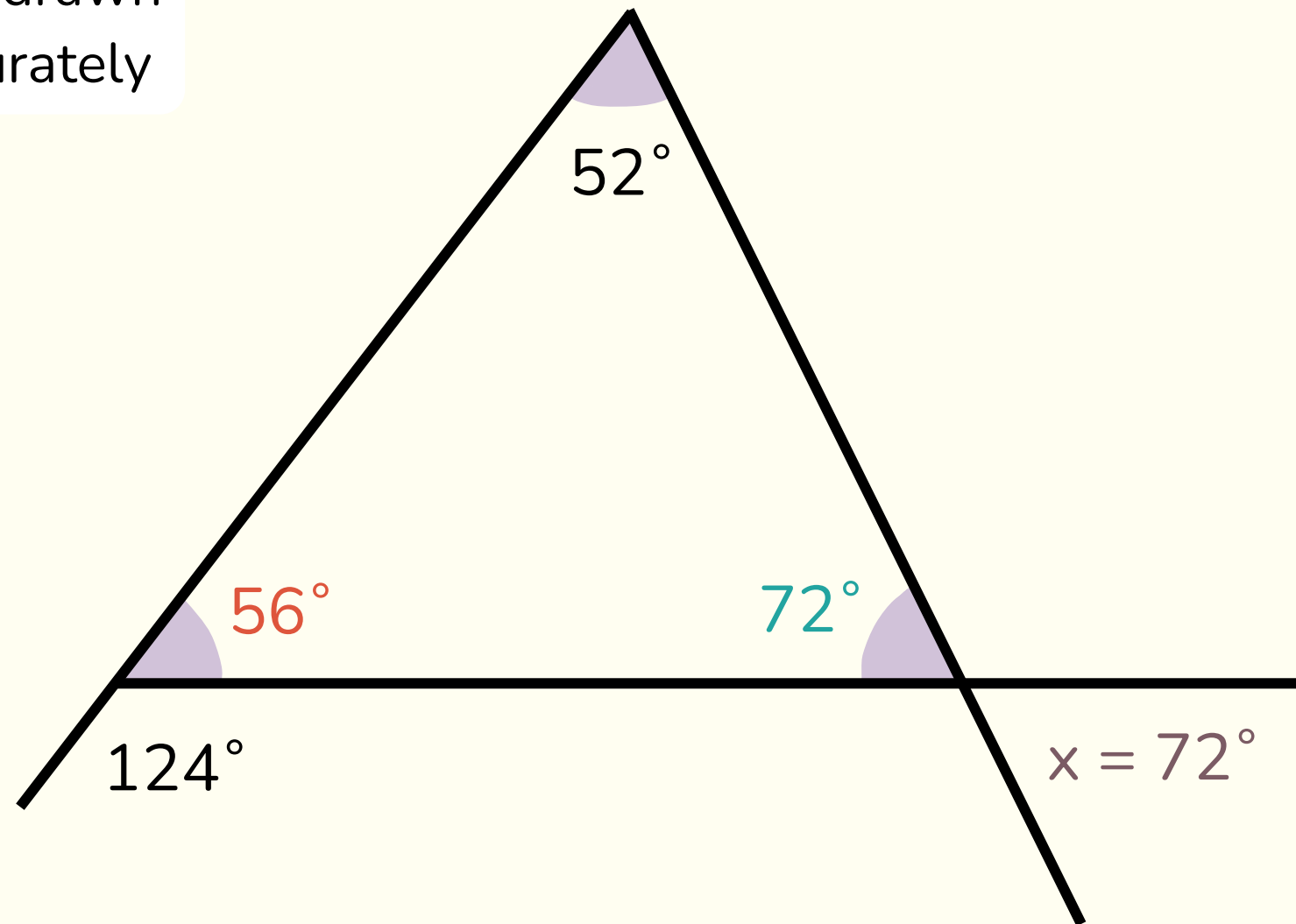
Find the missing angle  $x$ :

Not drawn  
accurately



# COMPLEX ANGLE PROBLEMS

Not drawn  
accurately



Find the missing angle  $x$ :

$$180 - 124 = 56^\circ$$

Adjacent angles on a straight line sum to  $180^\circ$

$$56 + 52 = 108^\circ$$

$$180 - 108 = 72^\circ$$

Angles in a triangle sum to  $180^\circ$

$$x = 72^\circ$$

Vertically opposite angles are equal

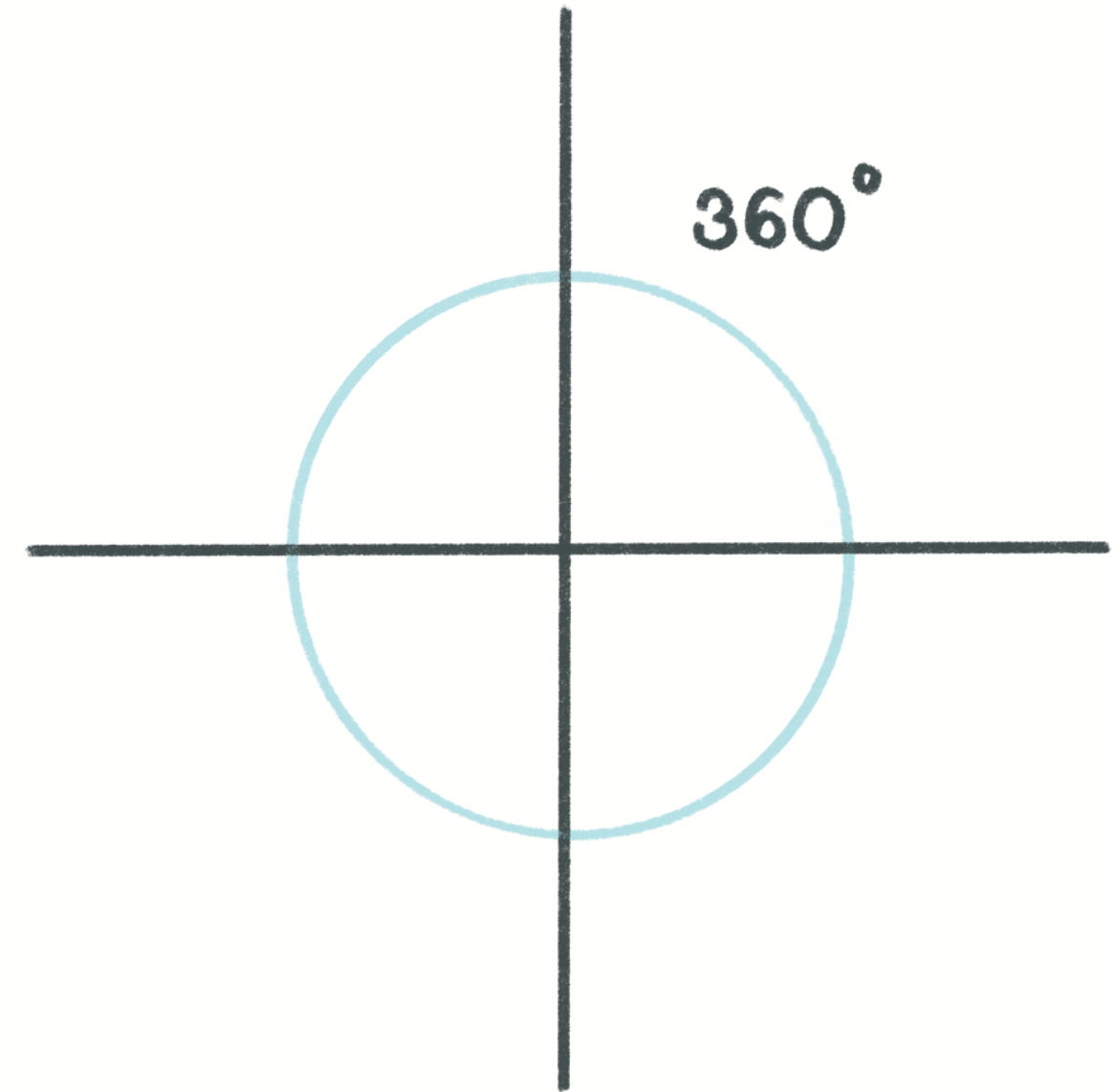


# MEASURING ANGLES





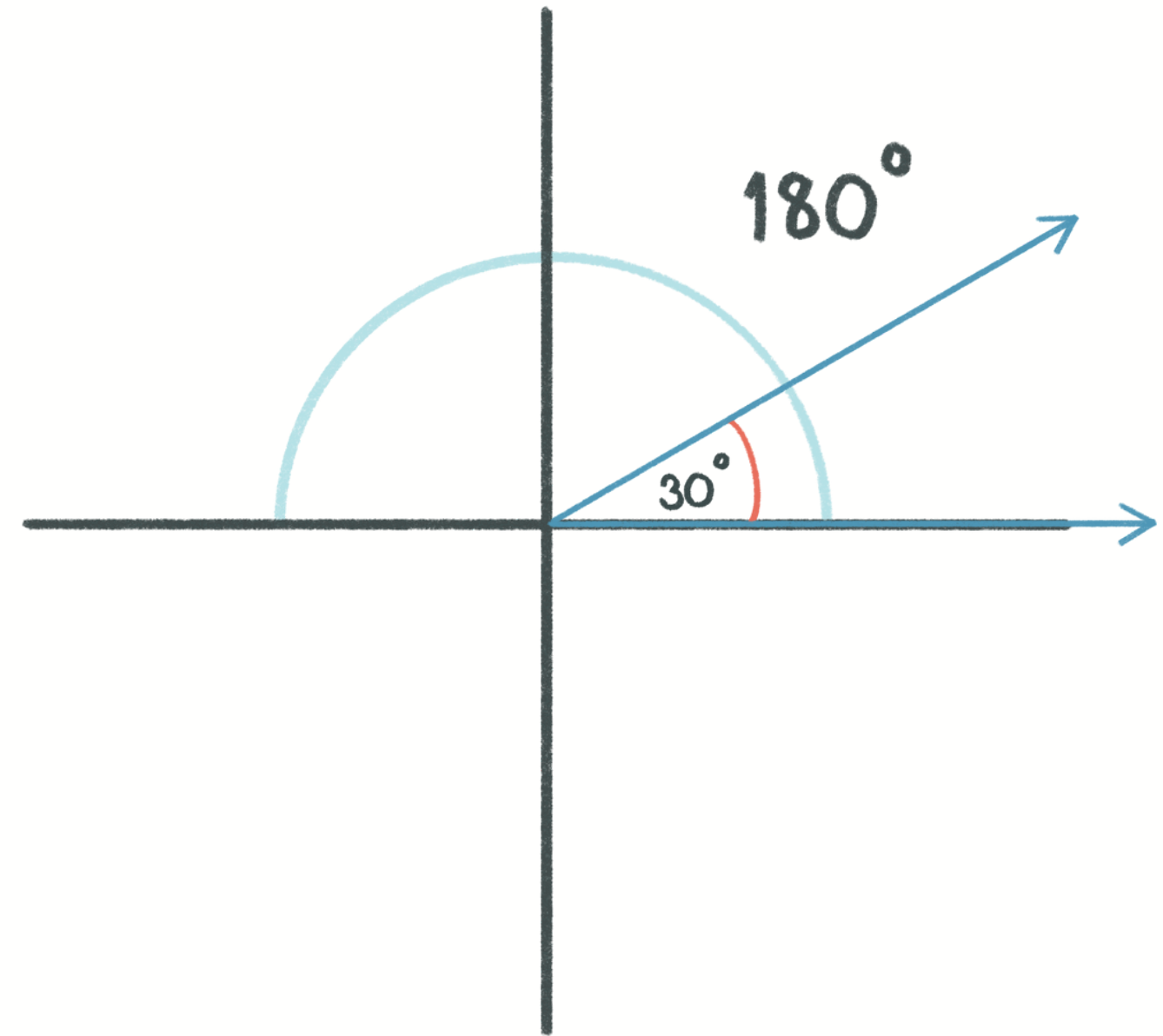
**An angle is measured with reference to a circle with its center at the common endpoint of the rays. A circle equals  $360^\circ$ . Each one degree that is used to measure is a fraction of this.**



If you are given an angle of  $30^\circ$  you can find the missing angle (n) using addition.

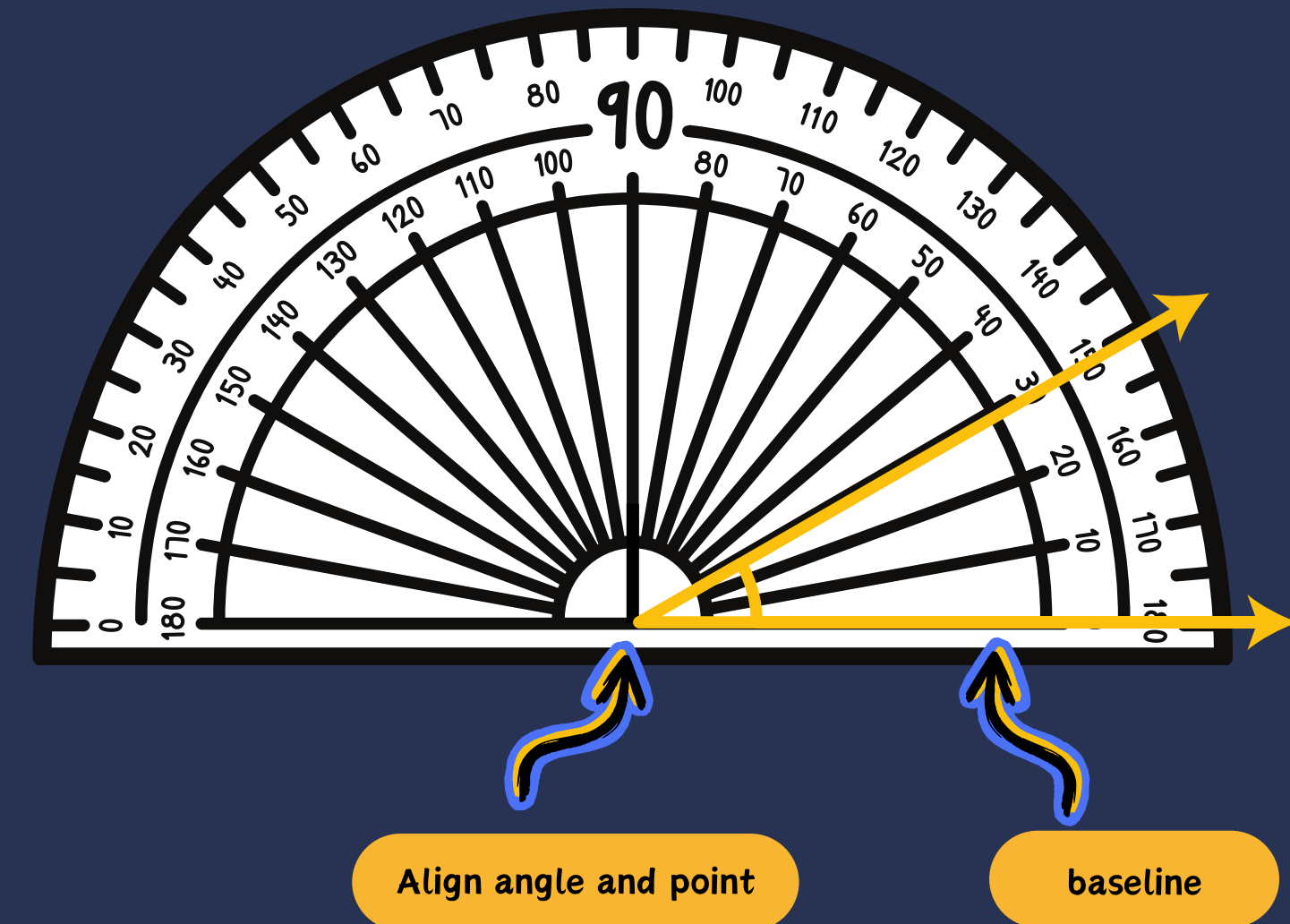
$$30^\circ + n = 180^\circ$$

$$n = 150^\circ$$



# How to use a protractor?

- 1 Place the Protractor: Align the baseline of the protractor with one side of the angle, ensuring the midpoint (center hole) is on the vertex of the angle.
- 2 Align the Baseline: Make sure one side of the angle follows the baseline (0° mark) of the protractor.
- 3 Read the Measurement: Look at where the other side of the angle intersects the numbered scale on the protractor. This is the measure of the angle in degrees.







# Quiz Time!

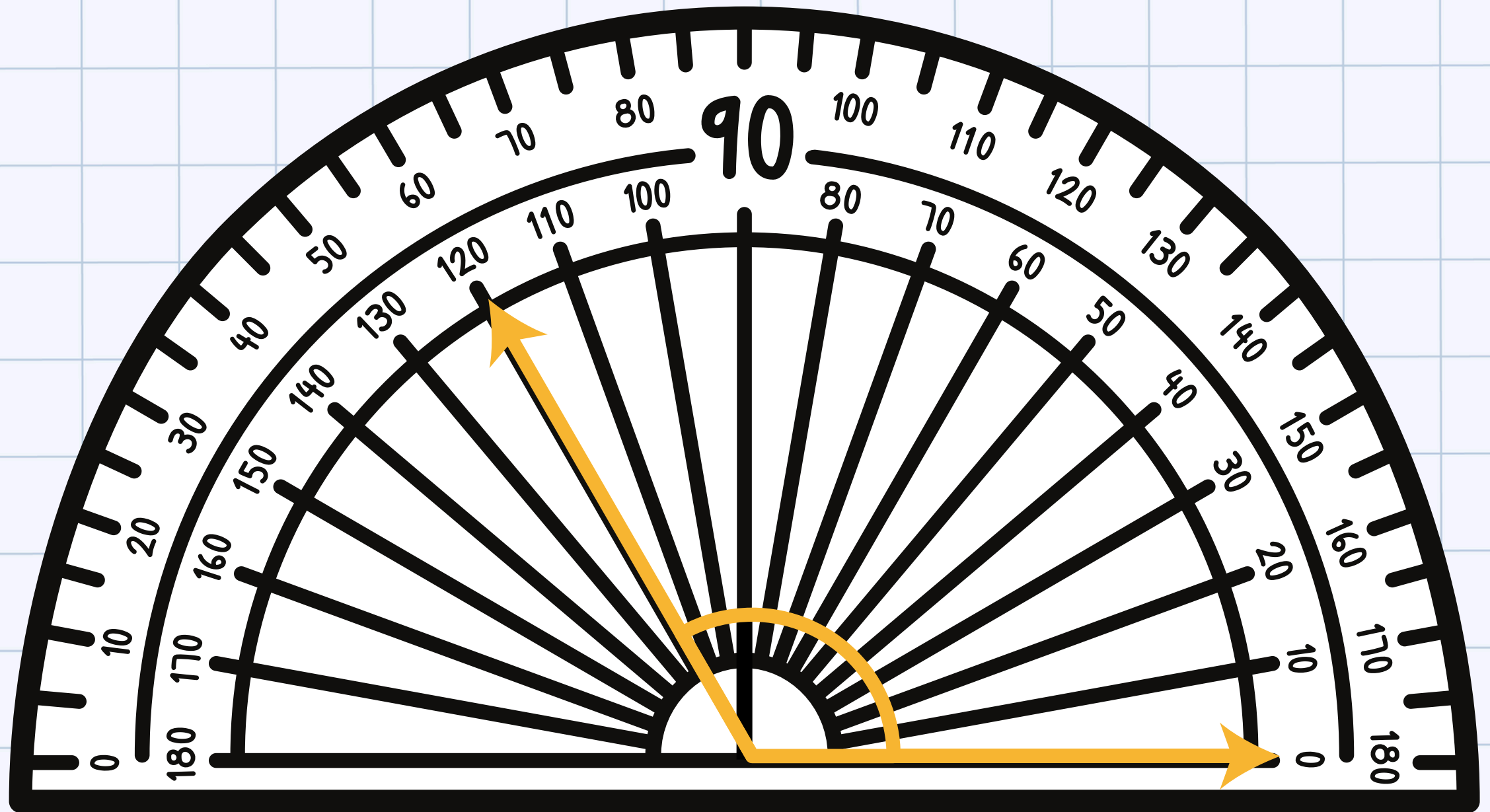


# What is the measurement of this angle?



A)  $120^\circ$

B)  $60^\circ$

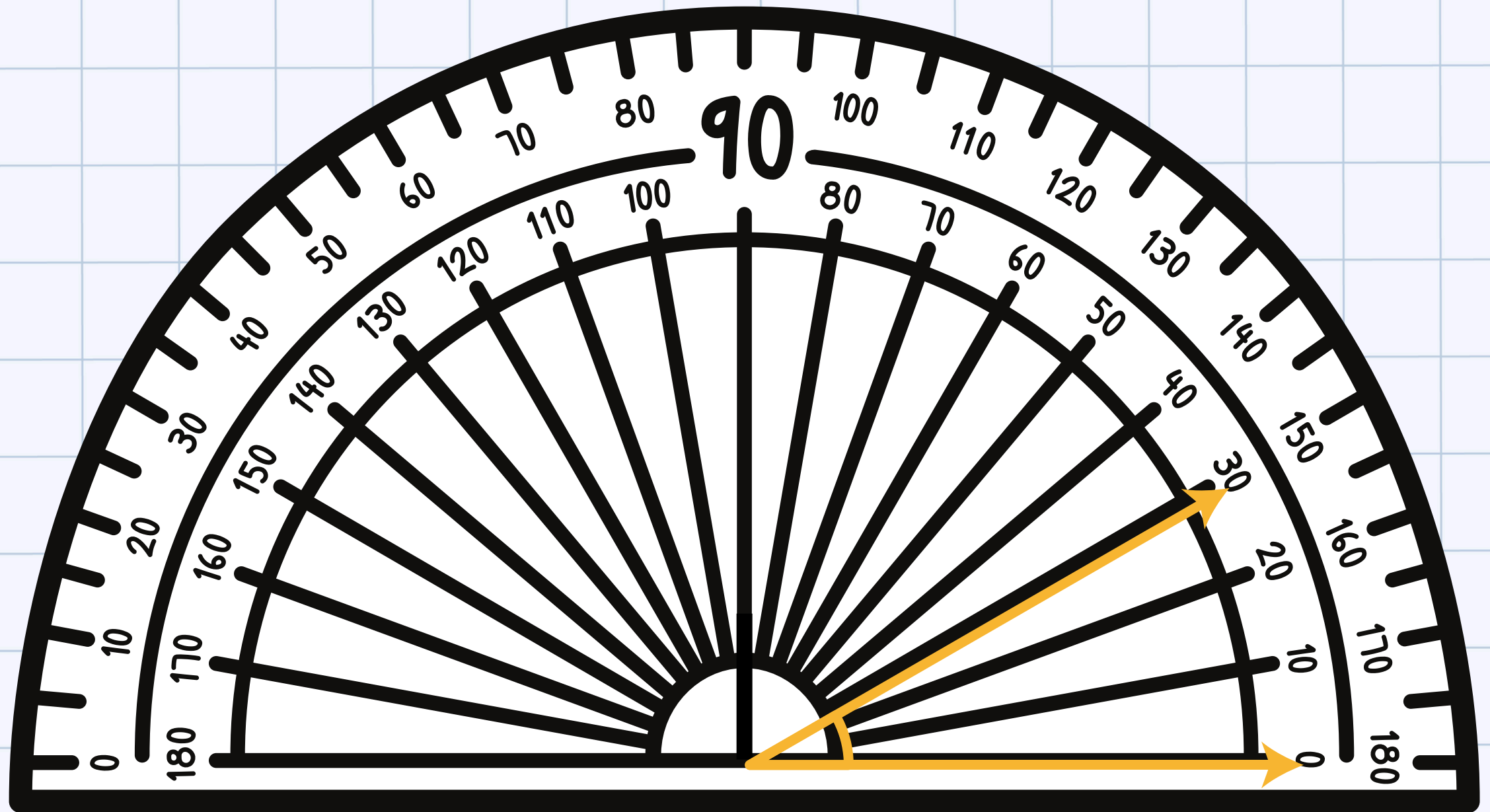


# What is the measurement of this angle?



A)  $150^\circ$

B)  $30^\circ$

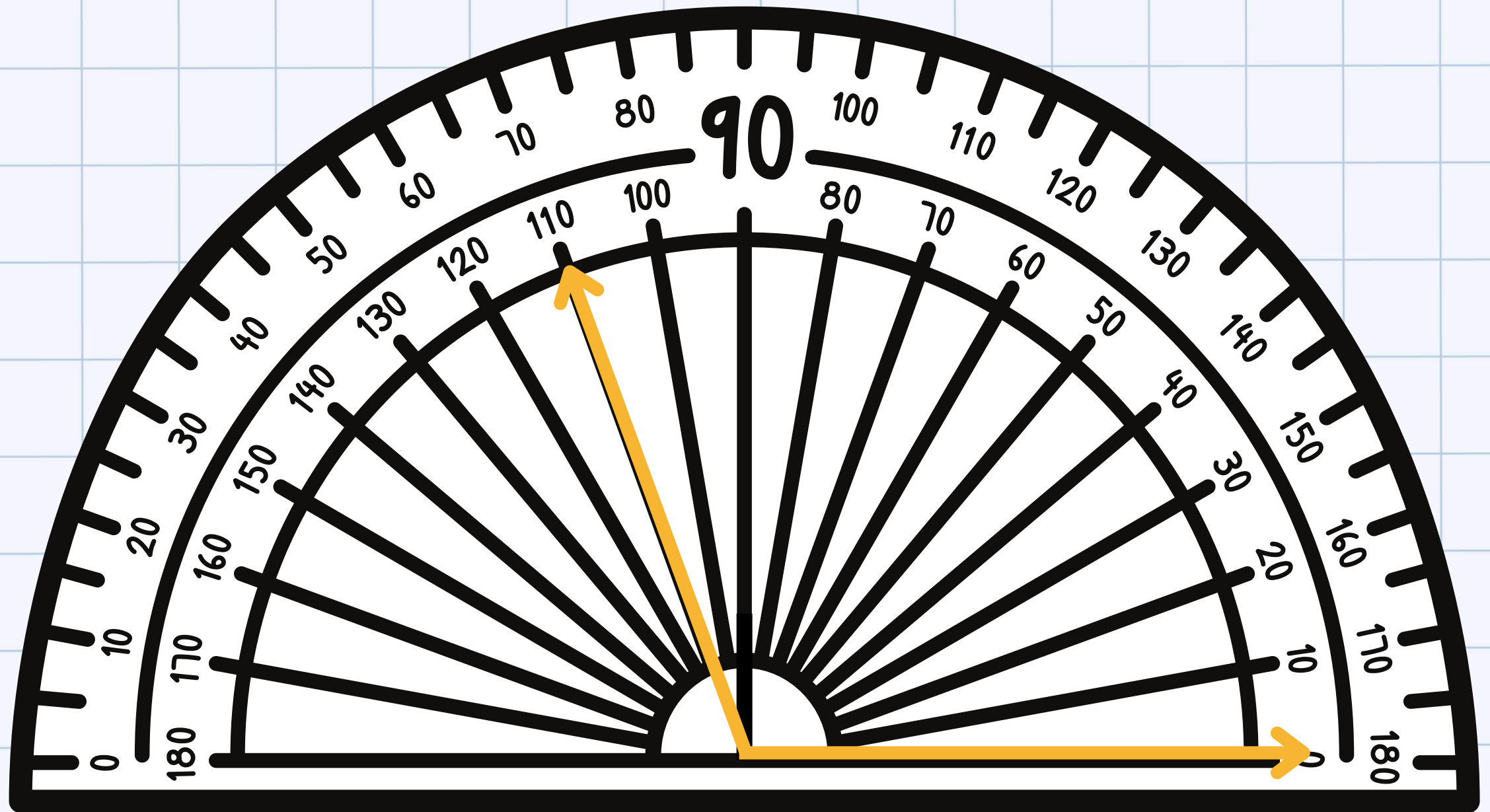


# What is the measurement of this angle?



A)  $110^{\circ}$

B)  $70^{\circ}$

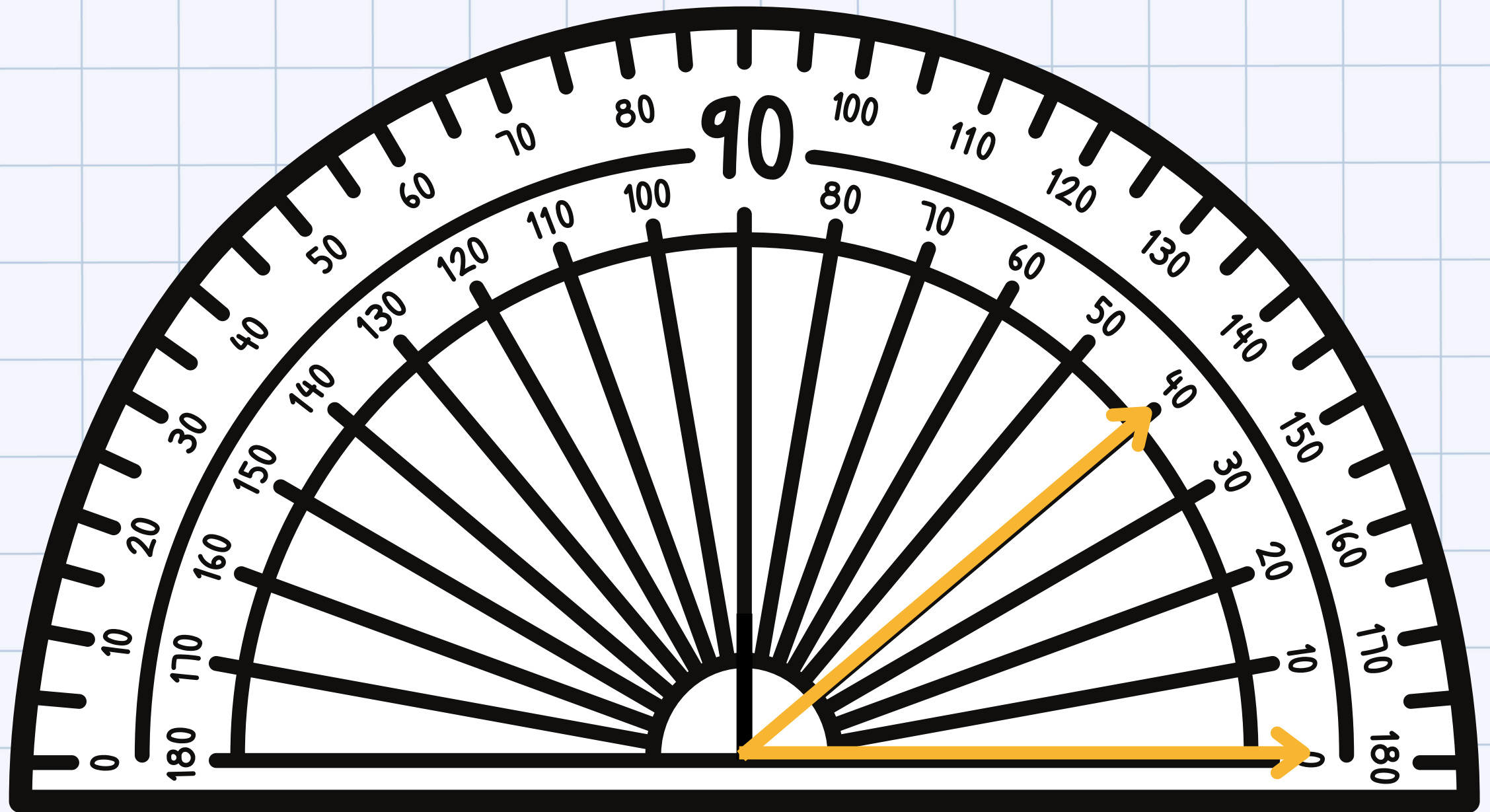


# What is the measurement of this angle?



A)  $140^\circ$

B)  $40^\circ$



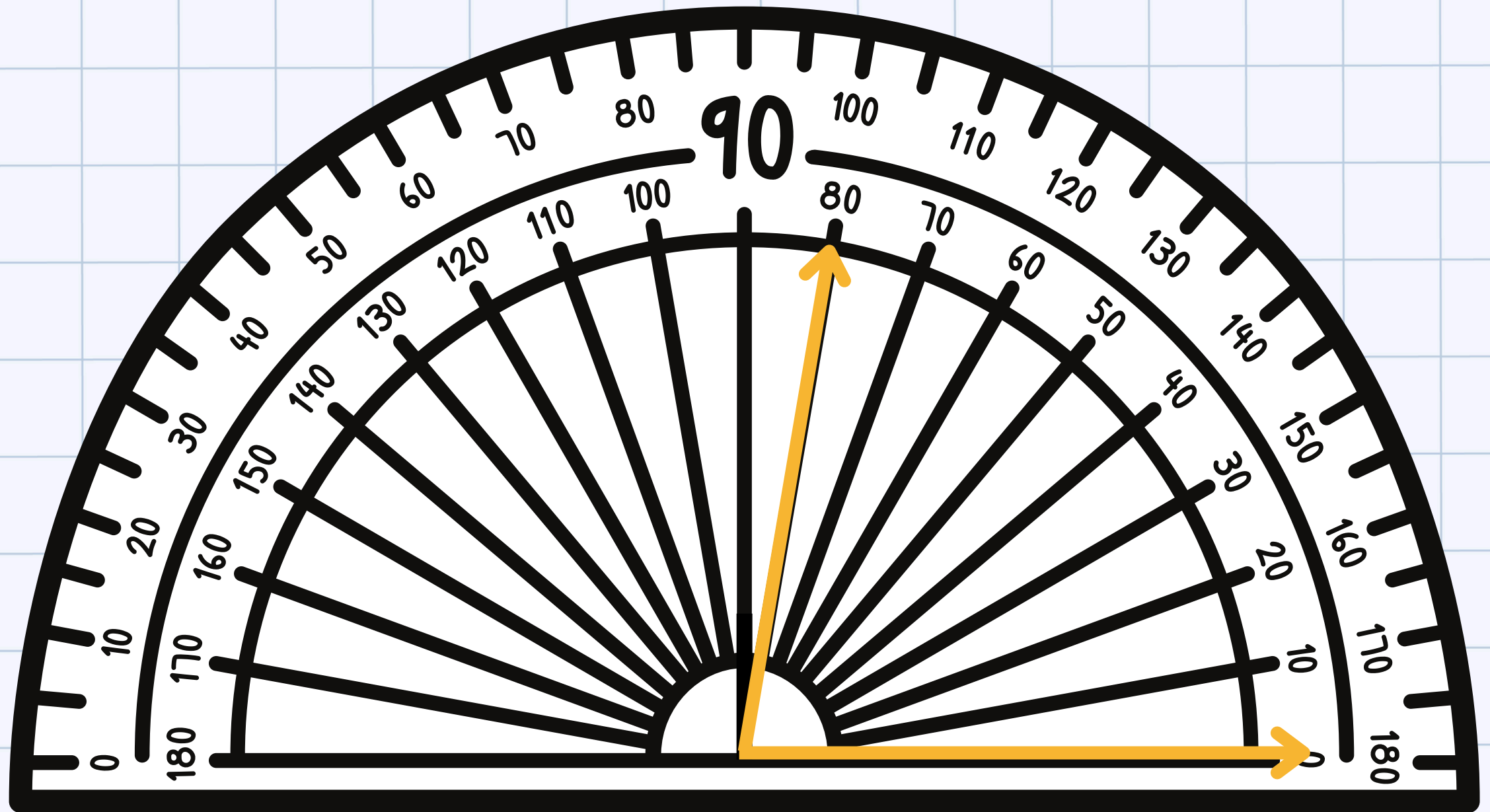


# What is the measurement of this angle?



A)  $100^\circ$

B)  $80^\circ$



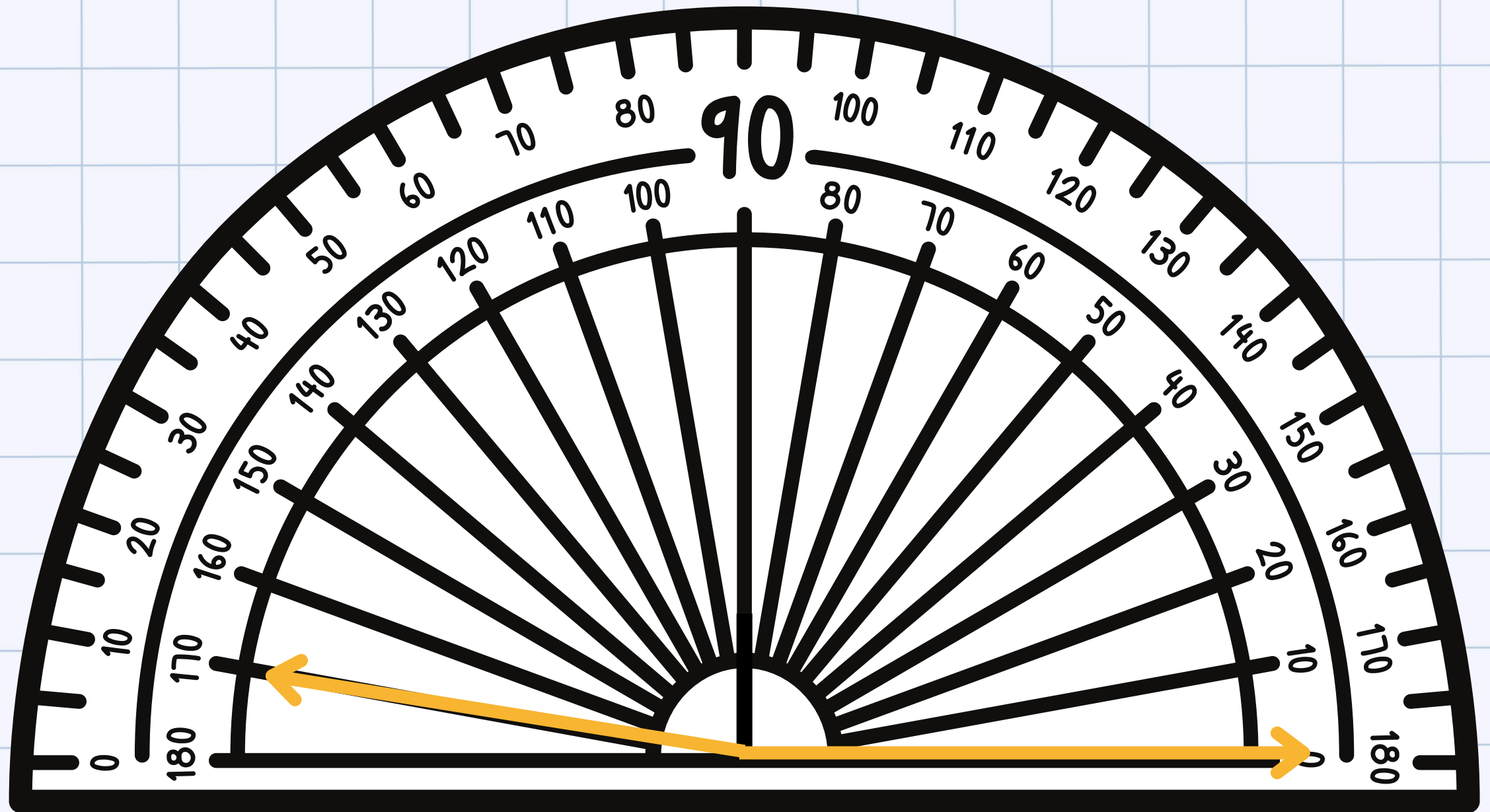


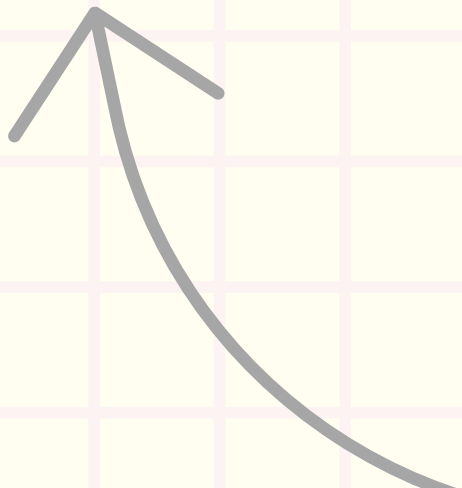
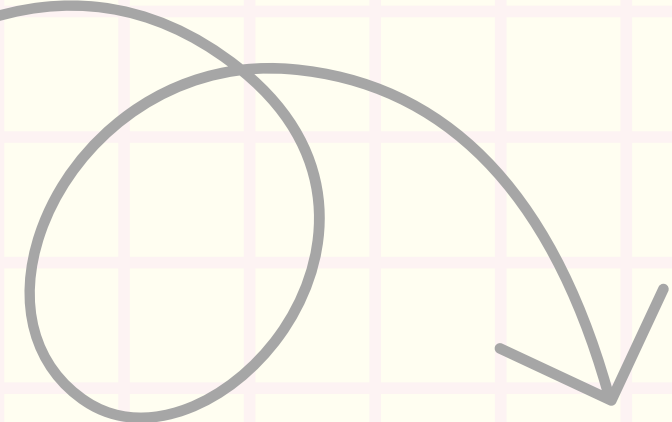
# What is the measurement of this angle?



A)  $170^\circ$

B)  $10^\circ$





**ANY QUESTIONS**